

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS)

CertifyX

Digital Certificate Generation & Verification Portal

► **Version:** 1.0

► **Domain:** Vibe Coding & Full Stack Development

1. Introduction

CertifyX is a web-based Digital Certificate Generation and Verification Portal developed to automate the complete certificate lifecycle in educational institutions. Instead of relying on paper-based processes, students can request certificates online while staff review, approve, and generate digitally signed certificates. The system stores every certificate securely and enables instant verification through QR codes and unique certificate IDs.

The solution improves operational efficiency, reduces paperwork, prevents certificate forgery, and offers a centralized digital repository for certificate management.

2. Objectives

- Digitize certificate request and approval processes.
- Reduce manual work and processing time.
- Provide secure QR-code based verification.
- Maintain a centralized certificate vault.
- Improve transparency and user experience.
- Generate reports and analytics.

3. User Roles

→Student

- ▶ Login and manage profile.
- ▶ Request Bonafide, Internship, Participation, Achievement, Conduct, Fee Paid, Hostel, NOC and other certificates.
- ▶ Fill request forms and upload supporting documents.
- ▶ Track request status.
- ▶ Download, share and verify certificates.

→Staff

- ▶ Review certificate requests.
- ▶ Approve or reject applications.
- ▶ Generate certificates with QR code and digital signature.
- ▶ Verify uploaded certificates.
- ▶ Manage events and reports.

→Admin

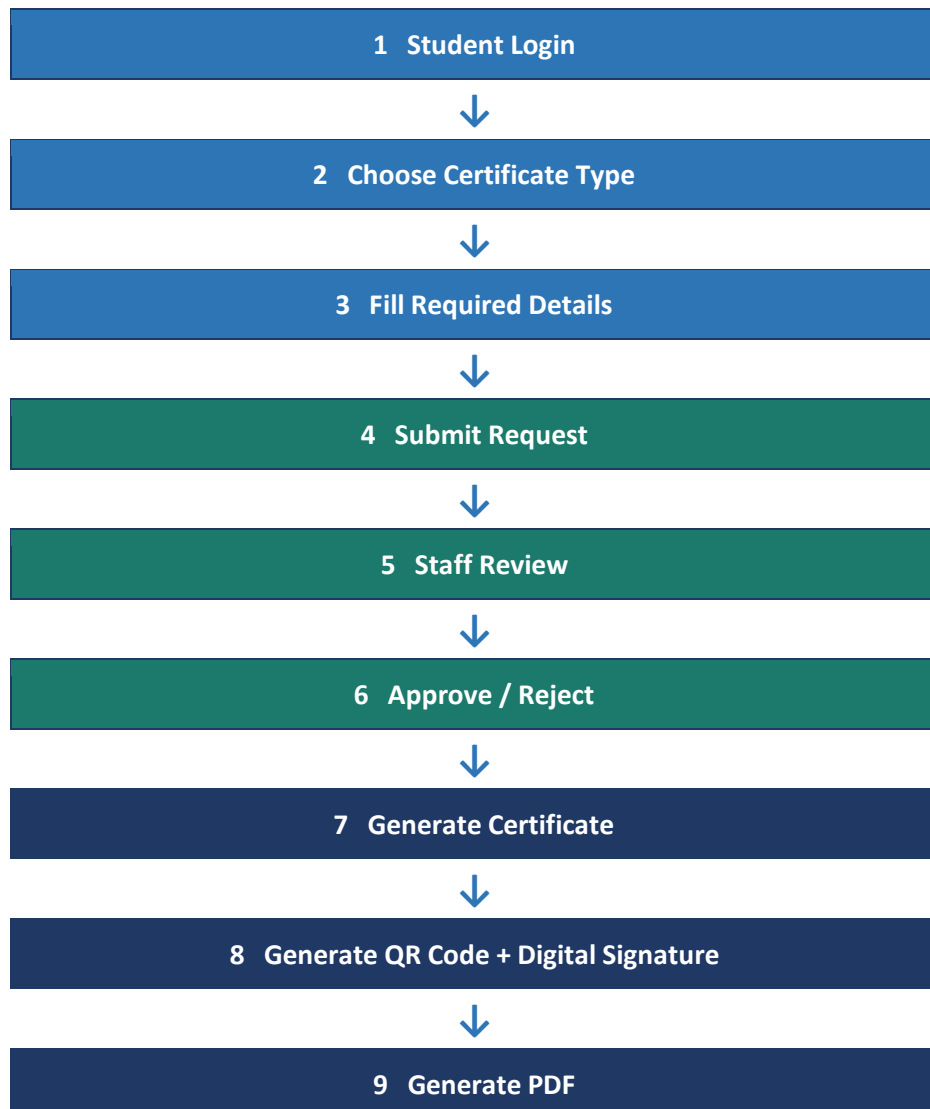
- ▶ View dashboard analytics.

- ▶ Manage staff accounts.
- ▶ Manage certificate templates.
- ▶ View reports and activity logs.
- ▶ Configure institution settings and notifications.

4. Functional Requirements

- **Authentication:** Secure login with role-based access.
- **Certificate Request:** Students select certificate type, enter required information and submit requests.
- **Approval Workflow:** Staff reviews each request and approves or rejects it with remarks.
- **Certificate Generation:** Approved requests automatically generate a PDF certificate containing the institution logo, digital signature, Certificate ID and QR code.
- **Certificate Vault:** Every generated certificate is stored in the student's personal repository for future access.
- **Verification:** Anyone can verify a certificate using the QR code or Certificate ID.
- **Notifications:** Automatic alerts are sent for request submission, approval, rejection and certificate generation.

5. System Workflow





10 Store in Certificate Vault



11 Student Downloads Certificate

6. Non-Functional Requirements

- **Security:** JWT authentication, encrypted passwords, role-based access, secure file storage.
- **Performance:** Fast response time and efficient certificate generation.
- **Reliability:** Data backup, activity logging and error handling.
- **Usability:** Responsive interface supporting desktop, tablet and mobile devices.
- **Scalability:** Supports multiple departments and large numbers of certificates.

7. Technology Stack

Layer	Technology
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB
Authentication	JWT
PDF Generation	PDFKit / pdf-lib
QR Code	QRCode Library
Email Notifications	Nodemailer

8. Advantages

- Paperless certificate management
- Faster approval workflow
- Secure verification
- Reduced human errors
- Centralized storage
- Easy access for students
- Improved institutional efficiency

9. Future Enhancements

- Blockchain-backed verification
- Mobile application
- ERP integration
- PKI-based digital signatures
- Cloud backup and synchronization

10. Conclusion

CertifyX provides an efficient and secure platform for managing certificate requests, approvals, generation and verification. By digitizing the complete workflow, the system improves transparency, minimizes manual effort and ensures authenticity through QR-based verification and digitally signed certificates.